INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

BEEF IN BARBECUE SAUCE, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers beef in barbecue sauce thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Tomato Paste

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0709, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

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(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Institutional Meat Purchase Specifications for Beef, Series 100

(Copies are available from the Director, Livestock and Meat Standardization Branch, Livestock and Seed Division, Agricultural Marketing Service, Room 2603, South Building, P.O. Box 96456, U.S. Department of Agriculture, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations (9 CFR Parts 300-399)

Food Safety and Inspection Services, Canning of Meat and Poultry Products (9 CFR Part 318)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

United States Standards for Condition of Food Containers

(Copies are available from the Chairman, Condition of Container Committee, Agricultural Marketing Service, Room 2506, South Building, P.O. Box 96456, U.S. Department of Agriculture, Washington, DC 20090-6456.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPS) Cuts: Item Number 114 (shoulder clod), 161 (round, shank-off, boneless), 163 (round, shank-off, 3-way, boneless), 165 (round, rump and shank-off, boneless), 167A (knuckle, peeled), 169 (top round), 170A (bottom round, heel-out), and 171B (outside round). All beef shipped between plants shall be accompanied by a USDA, Agricultural Marketing Service (AMS) Certificate to certify condition of the product and compliance with the IMPS Quality Assurance Provisions.

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3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days after initial certification is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 120 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence
- held after storage at an internal temperature not to exceed 40°F when further processing is resumed
- not refrozen after it has been tempered/thawed.

NOTE: Microwave tempering/thawing of the frozen beef is permitted provided that the beef is maintained in excellent condition, as determined by the USDA.

3.2.1.2 Beef, cooked, frozen. Frozen cooked beef shall be prepared from beef that complies with 3.2.1, and 3.2.1.1. The beef shall be processed in accordance with 3.3.1. The cooked beef shall be packaged in a USDA approved food packaging material and shall be frozen to 0°F or below within 72 hours. The beef shipped between plants shall be accompanied by a certificate of conformance (see 4.5.1.1) to certify the initial freezing (in-storage) date and compliance with the above stated requirements. The packaged, frozen, cooked beef shall have been held at 0°F or below for a period not to exceed 60 days prior to grinding (see 3.3.1).

3.2.2 Tomato paste. Tomato paste shall be U.S. Grade A or better of the U.S. Standards for Grades of Tomato Paste. The soluble solids content shall be as specified in 3.3.2.

3.2.3 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.4 Sugar, brown, light. The brown sugar shall be partially refined cane or beet sugar. It shall be light brown in color and shall have a sweet, molasses-like flavor.

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3.2.5 Vinegar, cider. Cider vinegar shall be prepared from apple cider, be clear, light amber in color, and free of haze, sediment, suspended matter, and floating particles. The cider vinegar shall contain not less than 4.0 percent and not more than 5.0 percent acetic acid.

3.2.6 Fat, beef. Beef fat shall be stabilized edible beef tallow with the following characteristics: Moisture: 0.2 percent maximum; Free fatty acid: 0.5 percent maximum as oleic; Color: slight yellow; Melting point: 110-112°F; AOM: 20 hours. The fat shall be maintained at 35°F or lower and held for no more than 30 days from time of rendering.

3.2.7 Starch, waxy maize, modified. The starch shall be a white, odorless, finely pulverized, modified waxy maize starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.8 Salt. Salt shall be noniodized, clear, white, refined, sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.9 Sauce, Worcestershire. The sauce shall be a brown or dark brown colored liquid having a pleasant, tart, peppery, fruit-spice flavor and shall possess the body and character typical of this sauce. The sauce shall have a titratable acidity (as acetic acid) of not less than 2.8 nor more than 3.8 percent.

3.2.10 Flavoring, smoke. Smoke flavoring shall be an aqueous solution of natural maple and/or oak wood smoke and shall be dark brown-black in color with a clean smoke odor. The smoke flavoring shall have a titratable acidity of 6.2 to 7.2 percent expressed as acetic acid, carbonyl level (butanone-2) of 7.0 to 8.0 grams per 100 mL and a phenolic level (dimethoxphenol) of 7.5 to 9.0 mg per mL.

3.2.11 Onion, dehydrated, chopped. The dehydrated chopped onion shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.12 Mustard flour. Mustard flour shall be a bright yellow powder prepared from a blend derived from the endosperm of the seed of Brassica hirta Moench and Brassica juncea. Mustard flour shall contain not less than 0.4 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.13 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

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3.2.14 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.15 Pepper, red, ground. Ground red pepper shall be derived from red, ripe fruit of Capsicum frutescens L. and shall possess the characteristic yellowish-red to red color with a perceptible sensation of pungency. It shall be uniformly ground to allow a minimum of 95 percent by weight to pass through a U.S. Standard No. 40 sieve, and not less than 95 percent by weight to be retained on a U.S. Standard No. 60 sieve. The Scoville Pungency Value shall be not less than 30,000 units.

3.2.16 Clove, ground. Ground clove shall be prepared from the dried, unopened flower buds of Caryophyllus aromaticus L. The powder shall be dark reddish-brown in color and shall possess a strong aromatic odor with a hot pungent taste. The clove shall contain not less than 15.0 mL of volatile oil per 100 grams of ground clove and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.17 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall have a fragrant clove-like aroma, a strongly aromatic pungent clove-like flavor, and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

3.2.18 Cinnamon, ground. Ground cinnamon shall be the dried bark of Cinnamomum burmanii Blume, commonly know as "Korintje Cinnamon." The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.19 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. The beef shall be prepared as follows:

a. Boned and trimmed meat meeting the requirements of 3.2.1 shall be further trimmed, if necessary, to ensure compliance with finished product requirements (see 3.6).

b. The boned and trimmed meat cuts shall be cooked by a commercially acceptable cook method to provide a product meeting the finished product requirements (see 6.4).

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c. The cooked meat cuts shall be cooled to an internal temperature of 28° to 40°F within 24 hours following the cooking process and held for not more than 48 hours prior to grinding. Alternatively, if the cooked meat cuts are to be held longer than 48 hours after cooking, they shall be placed in a freezer within 4 hours after cooking, and frozen to an internal temperature of 0°F or lower within 24 hours after being placed in the freezer. If the meat cuts are to be frozen, the meat shall not have been previously frozen and the meat shall be placed into separate oxygen impermeable casing materials approved by the USDA and FDA, and securely closed. The frozen, cooked meat shall be held at 0°F for a period of not more than 60 days prior to grinding. The frozen meat shall be tempered to an internal temperature of 28° to 40°F prior to grinding.

d. The tempered meat shall be mechanically ground through a grinder equipped with a 2-blade knife and a plate having openings measuring 1 inch in diameter.

3.3.2 Preparation of the sauce. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Tomato paste (24 percent solids) 1/	36.00
Water	30.20
Sugar, brown, light	11.00
Vinegar, cider (50 grain)	8.30
Beef fat, rendered	6.00
Starch, waxy maize, modified	2.50
Salt 2/	1.50
Sauce, Worcestershire	1.50
Flavoring, smoke	0.50
Onion, dehydrated, chopped	0.95
Mustard, flour	0.50
Garlic powder	0.50
Lecithin	0.30
Red pepper, ground	0.10
Clove, ground	0.10
Allspice, ground	0.03
Cinnamon, ground	0.02

1/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solids level to that specified and water shall be adjusted accordingly.

2/ The total amount of salt in the sauce shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

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NOTE: The following sauce preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. (When alternative procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. A slurry shall be made with the starch and part of the water. The remaining ingredients shall be mixed together and heated to approximately 140°F. The starch slurry shall be added and the final mixture heated to not less than 180°F but not more than 190°F and held for approximately 5 minutes at this temperature.

b. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating.

c. The sauce shall be used within 4 hours after cooking and held at a temperature of not less than 150°F during this interval.

3.3.3 Preparation of beef and barbecue sauce. The ground beef shall be mixed with the sauce only to the extent necessary to form a uniform mixture of meat pieces and sauce.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each tray pack can shall be filled with 106 ounces of product.

b. Immediately after filling each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirements (see 3.6s).

c. The filled and sealed tray pack can shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

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c. There shall be no color foreign to the product.

d. Total weight of cartilage, coarse connective tissue, sections of tendons or ligaments, and glandular material, collectively, in any individual can shall not be greater than 2 ounces.

e. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. The texture of the beef shall not be dry, tough, rubbery, or mushy.

g. Sauce consistency shall be smooth.

h. There shall be a uniform mixture of beef and sauce.

i. The average net weight shall be not less than 106 ounces.

j. No individual can shall have a net weight of less than 104 ounces of product.

k. Drained weight of beef pieces on a U.S. Standard 1/4 inch sieve in any individual tray pack can shall be not less than 44.0 ounces.

l. The average drained weight shall be not less than 48.0 ounces on a U.S. Standard 1/4 inch sieve.

m. Drained weight of beef pieces on a U.S. Standard 1/4 inch and a No. 7 sieve combined in any individual tray pack can shall be not less than 53.0 ounces.

n. The average drained weight shall be not less than 57.0 ounces on a U.S. Standard 1/4 inch and a No. 7 sieve combined.

o. The average fat content of the finished product shall be not greater than 8.0 percent.

p. No individual tray pack shall have a fat content greater than 10.0 percent.

q. No individual tray pack shall have a salt content greater than 1.5 percent or less than 1.0 percent.

r. The product shall show no evidence of excessive heating (materially darkened or scorched).

s. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

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3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and

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compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average total can content weight or the total can content weight of any individual can is less than specified in 3.6, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The sample cans shall be selected at random from the lot. The sample cans shall be heated in accordance with heating instructions on the can label, opened, and inspected for the defects listed in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	-	0	2
	8	16	1	2
3201 to 35000	13	-	0	3
	13	26	3	4

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1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and, if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Drained weight of beef pieces on a U.S. Standard 1/4 inch sieve in any individual tray pack can is less than 44.0 ounces 4/ 5/
102	Drained weight of beef pieces on a U.S. Standard 1/4 inch and a No. 7 sieve combined in any individual tray pack can is less than 53.0 ounces 4/ 6/
103	Texture of beef dry, tough, rubbery, or mushy
104	Total weight of cartilage, coarse connective tissue, sections of tendons or ligaments, and glandular material, collectively, in a can is greater than 2 ounces
105	Presence of a bone piece measuring 0.3 inch or more in any dimension
106	Sauce is lumpy or is excessively thick or thin
107	Not a uniform mixture of beef and sauce
108	Product shows evidence of excessive heating (materially darkened or scorched)

1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

3/ Interior of filled and sealed cans shall be examined visually for coating defects during product examination. Suspected rust spots shall be verified by testing in accordance with 4.5.7. Any spot verified as a rust spot shall be cause for rejection of the lot. Any scratch or fracture that penetrates through the coating of a can shall be cause for rejection of the lot.

4/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat without breaking the beef pieces. A U.S. Standard 1/4 inch sieve shall be placed on top of a U.S. Standard No. 7 sieve. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the beef pieces. The stack of sieves shall be tilted at approximately a 45° angle and allowed to drain for 2 minutes. The drained weight of the beef retained on each sieve shall be determined separately by subtracting the sieve tare weight of each sieve from the gross weight of each sieve. The drained weight shall be reported to the nearest 0.1 ounce.

5/ The lot shall be rejected if the sample average drained weight on a U.S. Standard 1/4 inch sieve is less than 48.0 ounces.

6/ The lot shall be rejected if the sample average drained weight on a U.S. Standard 1/4 inch and No. 7 sieve combined is less than 57.0 ounces.

4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The tray pack cans shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat and salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with 4.5.4.1. In addition, the following defect shall be classified as a major defect.

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Evidence of buffing with an abrasive substance (see 5.1.1).

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, for example, label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closure shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and CFR 21, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to $75^{o} \pm 5^{o}F$, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. One measurement shall be made when examining a ribbed lid; lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Test for rust spots on interior of tray pack cans. Where rust spots are suspected on interior of tray pack cans, the following test shall be performed:

a. Immerse a cotton swab in acetone or methyl ethyl ketone solvent and gently rub suspected spot. Handling of the solvent shall be in strict accordance with the guidelines of the manufacturer's Material Safety Data Sheet (MSDS).

b. If suspected spot immediately disappears, it is not a rust spot.

c. If suspected spot remains, continue gently rubbing with the swab re-soaked with solvent, and observe for disappearances or retention of spot. Replenish solvent as necessary.

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d. If spot persists, and all but a very thin interior can coating or no interior can coating remains, the spot shall be scored as a rust spot. This can be confirmed further by rubbing with a finger. The rust spot will be felt as a slight bump over the substrate.

4.5.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed, and thermoprocessed as specified in 3.4 and 3.5. The practice of reconditioning tray pack cans by buffing with an abrasive substance shall not be permitted (see 4.5.4). The interior coating of filled and sealed thermoprocessed cans shall be free of rust spots, scratches, or fractures that penetrate through the coating when examined in accordance with 4.5.3.2.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical.

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The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three-digit letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

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1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1990, would be coded as 0078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four-digit Julian code. Following the four-digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.5.4.1 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

As an alternate labeling method, a preprinted, self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

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5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT
OR MORE THAN TWO HIGH IN STORAGE,
UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Ordering data. Acquistion documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Beef preparation. Satisfactory product has been produced by cooking the beef to a yield of 70 to 75 percent at a maximum beef temperature of 155°F and using an approximate fill ratio of 60 percent beef and 40 percent sauce. These percentages may be adjusted, if necessary, to assure compliance with finished product requirements.

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6.5 Subject term (key word) listing.

Canned foods
Combat field feeding
Food processing
Meat
Operational rations
Shelf stable

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8940-0707)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-44224B	2. DOCUMENT DATE (YYMMDD) 1990 September 20

3. DOCUMENT TITLE
BEEF IN BARBECUE SAUCE, THERMOSTABILIZED, TRAY PACK

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) (2)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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